

DLA26BZ03-NV011: Digital Twin of the Organization for Enhanced Mission Readiness

PROJECTED CMMC LEVEL REQUIREMENT:

Level 2 (Self)

TECHNOLOGY AREAS:

None

MODERNIZATION PRIORITIES:

Advanced Materials | Mission Readiness & Disaster Preparedness | Nuclear | Sustainment & Logistics

KEYWORDS:

Digital Twin, Workforce, Organization, Artificial Intelligence, Synthetic Data, Scenario Planning, Great Power Competition, Productivity, Logistics, Mission Readiness

ADDITIONAL INFORMATION

N/A

OBJECTIVE:

The Defense Logistics Agency (DLA) seeks to develop and implement a dynamic workforce digital twin to enhance organizational effectiveness, support mission-critical decisions, and accelerate the adoption of advanced technologies. As Artificial Intelligence (AI) fundamentally changes the workforce of the future, the Department must achieve a 10x productivity increase to maintain strategic overmatch and compete effectively with near-peer adversaries and the workforce plays a critical role in achieving this scale. The primary objective is to create a Digital Twin (DT) of the organization capable of generating synthetic data and enabling leaders to react quickly to dynamic changes. This tool will inform key workforce-related decisions, improve how work gets done, and allow for robust scenario-planning for core mission activities under unpredictable global conditions.

ITAR:

The technology within this topic is restricted under the International Traffic in Arms Regulation (ITAR), 22 CFR Parts 120-130, which controls the export and import of defense-related material and services, including export of sensitive technical data, or the Export Administration Regulation (EAR), 15 CFR Parts 730-774, which controls dual use items. Offerors must disclose any proposed use of foreign nationals (FNs), their country(ies) of origin, the type of visa or work permit possessed, and the statement of work (SOW) tasks intended for accomplishment by the FN(s) in accordance with section 3.5 of the Announcement. Offerors are advised foreign nationals proposed to perform on this topic may be restricted due to the technical data under US Export Control Laws.

DESCRIPTION:

A digital twin of the organization is a critical enabler for the DLA, empowering leaders to model and test workforce scenarios, optimize processes, and achieve the exponential efficiency gains required for modern great power competition. Current static models are insufficient for the agility needed today. The proposed solution will move beyond legacy systems by integrating real-time data to provide dynamic insights into workforce capabilities, productivity limitations, and the potential impacts of organizational restructuring. Crucially, this organizational DT must possess the capability to generate synthetic data to simulate complex, unforeseen scenarios, allowing the enterprise to react rapidly to supply chain disruptions, geopolitical shifts, or internal surges. By modeling the workforce of the future—one that is heavily augmented by AI—this effort will explore innovative data sources and functionalities to address critical questions:

- Strategic Competition & Productivity: How can we leverage the DT to identify structural and process pathways to a 10x productivity increase, countering the scale of adversaries like China?
- Surge Deployment & Rapid Reaction: How ready is the workforce for a sudden surge scenario? Utilizing synthetic data generation, how can we proactively test personnel moves to backfill deployed staff without mission degradation?

- **AI Integration & Organizational Structure:** What are the root causes of manual work, and what activities must be stopped or automated by AI? How does the organizational structure need to shift to maximize human-machine teaming and efficiently integrate new mission sets?

Research and Development (R&D) efforts for this topic should demonstrate a technical feasibility that has not been fully established and must be judged to be at a Technology and/or Manufacturing Readiness Level (TRL/MRL) between 3 and 6 to be considered for funding.

PHASE I:

Not to exceed a duration of 12 months and a cost of \$100,000. Phase I will focus on exploring the value of a digital workforce twin by delivering a proof-of-concept. This phase includes:

1. Identifying and ranking high-impact use cases (e.g., surge deployment in human capital, order-to-cash in finance, human-machine teaming integration).
2. Identifying necessary data sources, articulating a data acquisition and synthetic data generation plan, and defining success criteria.
3. Developing a business case on the expected ROI of the tool and a product roadmap for a Minimum Viable Product (MVP).

An alignment or collaboration with a relevant DLA Component organization (e.g., DLA J1, J6, J3, and J7) is highly desirable.

PHASE II:

Phase II will consist of establishing a pilot-scale process to build, test, and validate a functioning Minimum Viable Product (MVP) of the digital workforce twin solution. This will be conducted in a series of milestone-driven agile sprints to rapidly deliver incremental value and incorporate user feedback. The goal is to produce a functioning tool that can simulate scenarios via synthetic data, scale to other use cases, and refine the business case for further investment based on demonstrated mission value in increasing productivity. Collaboration with a relevant DoW Component organization is highly desirable.

PHASE III DUAL USE APPLICATIONS:

At this point, no specific funding is associated with Phase III. The progress and relationships developed in Phases I and II should position the solution for transition to production in support of DoW orders. The underlying technology can be adapted for a wide range of commercial and public sector applications, including corporate strategic planning, human resource management in large enterprises, and organizational design for complex, non-DoW government agencies navigating the AI transition. The goal is the organic growth of the business by applying the solution to both defense and commercial markets.

REFERENCES:

1. Digital Twins – when and why to use one
2. What is Digital Twin technology?
3. Digital twins: How to build the first twin

TECHNICAL POINT OF CONTACT (TPOC):

None